# The Artificial Neuron KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**1. Should I go?** Answer 1 (yes) or 0 (no). Then choose how much each one matters to you (0 to 3).

| | **Answer (1 or 0)** | **How much it matters** | **Answer $\times$ how much** |
|---|---|---|---|
| Is it sunny? | Answers vary. | | |
| Are my friends going? | | | |
| Is my homework done? | | | |
| | | **Total** | |

Is the total 4 or more? _____ Do I go? _____ Did my partner decide the same? _____

**2. Multiply, add, compare.**

An artificial neuron multiplies each input by its weight, adds them up, and compares the total with the threshold.

Inputs: 1, 0, 1. Weights: 3, 2, $-1$. Threshold: 2.

Total = $(1 \times 3) + (0 \times 2) + (1 \times -1) =$ 2 Does it fire (2 or more)? yes

Inputs: 1, 1, 1. Weights: 1, $-2$, $-1$. Threshold: 2.

Total = $1 - 2 - 1 = -2$ Does it fire? no

**3. Be a neuron.** Weights: **A:** +2, **B:** +2, **C:** $-1$. Threshold: **3 or more**.

| **Round** | **Inputs that are on** | **Total** | **Fire?** |
|---|---|---|---|
| 1 | A and B | 4 | yes |
| 2 | A and C | 1 | no |
| 3 | A, B, and C | 3 | yes |
| 4 | B and C | 1 | no |
| 5 | A only | 2 | no |

**4. A neuron that sees.** Write the weights that got the most test pictures right.

| | | |
|---|---|---|
| **0** | −1 | **0** |
| −1 | +3 | −1 |
| **0** | −1 | **0** |

Threshold: 0

Test pictures right: 6 out of 6

One good answer. Others work too.

Which squares got **positive** weights? Why? The center: it is black in an X, not in an O.

Which squares got **negative** weights? Why? The sides: they are black in an O, not in an X.

**5. Real neuron, artificial neuron.** Match each part.

| **Real neuron** | **Artificial neuron** |
|---|---|
| Dendrites take signals in | inputs |
| Strong or weak synapses | weights |
| Fires only if the total is big enough | the threshold |

**6. Talk about it.** ChatGPT has billions of weights. Nobody can choose them by hand.

I think an AI with billions of weights gets them by Answers vary. (Next lesson: it learns them.)